

Predicting the Annual Dirty Ten Based on Open Data about Pollution Releases

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Abstract: The German Pollutant Release and Transfer Register (PRTR) contains over 80,000 reports from more than 5,000 companies, offering substantial transparency regarding environmental pollution. This research uses PRTR data to predict the annual "Top Ten Dirty," which are industrial entities expected to report the ten largest releases of a specific substance. Following the CRISP-DM process model, an analytical dataset was constructed and used to train two machine learning models: one to predict the annual, branch-specific top ten polluters for greenhouse gas emissions into the air and another for heavy metal discharges into water. The results demonstrate that both models produce predictions closely aligned with actual pollution reports. These predictions can support various stakeholders, including corporate environmental managers, regulatory agencies, business organizations, and authorities combating environmental crimes.

Keywords: Corporate Pollution Reporting, Data Science, Machine Learning, Pollution Prediction, German PRTR register

1 Introduction

Historical pollution data from official registers, such as the European Pollutant Release and Transfer Register (E-PRTR), is crucial for many reasons, including environmental monitoring, regulatory purposes, and transparency. Equally important is the ability to predict future environmental impacts for stakeholders such as corporate sustainability managers, investors, regulatory bodies, communities, and individuals living close to industrial facilities. For example, accurate forecasts of greenhouse gas (GHG) emissions are essential for developing effective pollution reduction strategies [JC22]. This study explores using Machine Learning (ML) techniques to predict future polluters, focusing on identifying the top ten annual polluters by industry branch, also referred to as the 'annual dirty ten' companies [EE23]. These companies report the ten largest annual pollution amounts above the legal threshold for a particular type of pollution.

This analysis is based on the German PRTR dataset. Two Random Forest models were trained to predict the top ten polluters: one for GHG emissions into the air and another for heavy metal discharges into water.

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The study details the data preparation techniques and ML methods applied, following the CRISP-DM process model, which is widely used in Data Science and ML research and practice [Ma21]. Performance evaluation metrics demonstrate that the models' predictions closely align with actual pollution reports.

Existing prediction methods often rely on large datasets, such as Earth observation data and Internet of Things (IoT) sensor data. These methods typically focus on short-term predictions of air and water quality. These approaches require complex data management systems and significant computational resources. In contrast, this study uses structured PRTR data to train ML models that can make annual predictions without requiring extensive infrastructure or high computational costs.

The following sections present a review of related research, an overview of the research methodology and design, and a description of the German PRTR dataset. The subsequent section details the main steps of the approach to using ML to predict the top 10 polluters. The article ends with the researchers' conclusions.

2 Related Work

Numerous studies across different disciplines have employed data from national PRTRs or combined registers to explore industrial pollution and its impacts. For instance, an empirical financial analysis examined how firms' environmental disclosures influence market valuation and investments in pollution prevention using toxic release data from the Japanese PRTR [HM10]. Ultimately, the study found that the Japanese financial market does not adequately price the risks associated with toxic chemical releases.

In Sweden, researchers developed a methodology to estimate the national environmental impact of chemical emissions using PRTR data [SPF16]. They aggregated and transformed emission data for 54 substances released into the air and water in 2008 using a life-cycle impact assessment approach. This enabled them to calculate potential human toxicity and ecotoxicity at a national level.

Erhart et al. [Er25] investigated the exposure of freshwater biodiversity in Natura 2000 sites to industrial pollution by spatially linking protected areas with emission data from the E-PRTR. They found that many protected sites are subject to elevated ecological risks, highlighting the urgent need for integrated conservation and pollution control measures.

Other researchers have explored the potential of PRTR data to support cleaner production initiatives. One study examined an Australian fertilizer manufacturing facility to determine if PRTR reporting could help identify and implement cleaner production strategies [KS04]. While the study emphasized the utility of PRTR data for program development and monitoring, it also noted a key limitation: cleaner production programs typically require process-level pollution data, whereas PRTRs report emissions at the facility level.

A New Zealand study [HM10] used Australian PRTR data as a proxy for environmental performance and introduced a toxicity-weighting scheme for over 90 chemicals. The authors argued that toxicity-weighted indicators offer a more meaningful alternative to mass-based metrics for stakeholders, including corporate managers, policymakers, and analysts.

Erhart and Erhart [EE23] contributed a large-scale environmental ranking of over 10,000 European industrial facilities by integrating toxicity and global warming potentials using a harmonized life cycle assessment framework. Their results demonstrate substantial disparities in environmental performance across industries and regions, offering guidance for targeted policy interventions and corporate environmental improvements.

3 Research Methodology and Design

The study employed an experimental, quantitative research methodology and used historical PRTR data as the primary data source. This methodological approach resulted in a research design that ensures rigorous, reproducible, multi-step work, establishing a foundation for future research and improvements to predictive models. The steps are described in the remaining paragraphs.

Data collection. The dataset was obtained from the German PRTR register as of May 2023. It includes notifications from companies reporting their annual pollutant emissions and amounts of dangerous and non-dangerous waste across different pollution categories.

Data preparation. The collected data underwent preprocessing to handle missing values identified as 'NaN' (not a number), encode categorical variables, and normalize numerical features. Reports of non-targeted pollution types were retained, as they are relevant for identifying hidden patterns for the prediction task.

Exploratory Data Analysis (EDA) and Feature Selection. EDA was conducted to understand the distribution, patterns, and relationships within the dataset to help identify key variables influencing pollutant emissions. Descriptive statistics, visualizations, and correlation analyses were applied to assess data characteristics, detect outliers, and identify missing values.

Model development. Two binary classification models were developed using supervised ML techniques: one that predicts the top 10 polluters regarding GHG emissions into the air (AirT10P) and another that predicts the top 10 polluters regarding the release of heavy metals into the water (WaterT10P). These models predict whether a company falls within the top ten polluters based on reported emissions. Random forest classifiers [Gu22] were chosen for their interpretability, accuracy, and ability to cope with imbalanced datasets.

Data resampling. Since the dataset exhibited class imbalance, especially with fewer notifications belonging to the top 10 polluters, data resampling techniques were applied.

Downsampling was performed to reduce the number of non-top-10 cases to match the number of top-10 notifications. Upsampling was applied to increase the number of top-10 cases by duplicating existing data points, thus balancing the dataset.

Model training and validation. Two binary classification models were trained to predict whether a company would rank among the top 10 polluters (T10P): a model to predict top 10 polluters regarding GHG emissions into the air (AirT10P) and a prediction model regarding heavy metal emissions into water (WaterT10P). K-fold cross-validation [Ya23] with $k=5$ was used to evaluate model performance.

Secondary validation. To confirm the robustness of the models further, an additional validation step was performed. During resampling, notifications with NaN values for AirT10P and WaterT10P were removed from the training dataset. These notifications were then reintroduced as a separate validation dataset to test the models' ability to correctly classify instances outside the training set. The performance results were consistent with those of the initial validation step, with only minor differences at the second decimal place.

Model deployment and validation based on new data. To obtain insights into model deployment, the models were applied to predict the top ten polluters in terms of GHG emissions and heavy metal emissions for the year 2023. The results were then compared to the actual top ten polluters. The confusion matrix obtained from these comparisons revealed that the ML models provide highly accurate predictions. However, both models predicted the top 10 polluters with relatively low, yet still acceptable, precision.

4 The German PRTR Dataset

The German Federal Environmental Agency publishes the German PRTR dataset on its official website and updates it annually within the first two months of the calendar year. This study used a dataset downloaded in October 2024 that contained approximately 80,000 pollution reports from 2007 to 2022. These reports were submitted by approximately 5,000 industrial facilities, including small businesses, SMEs, and large multinational corporations. Over the 16-year period, 105,163 business activities were registered, 83,484 of which were designated as primary business activities.

Companies are classified annually according to the NACE Code standard [Eu22], which identifies their primary business activities. Each activity is also assigned a PRTR-specific code and categorized into one of nine industry branches: waste and wastewater management, the chemical industry, the energy sector, intensive live-stock farming and aquaculture, the food industry, the metal industry, the mineral processing industry, the paper and wood industry, and other industrial sectors.

The relevant key characteristics of the German PRTR dataset are as follows. Each company is assigned a unique identifier for each reporting year, along with updated information such as the company's name, address, and NACE code. An additional

identifier remains consistent across all reporting years. Each company's activities are described annually using PRTR-specific activity codes and categorized into industry branches. Reports classify pollution as hazardous or non-hazardous waste, wastewater shipments, or controlled or inadvertent releases into the air or water. For wastewater, air, and water releases, annual amounts of specific substances are reported and categorized by substance group, such as heavy metals and GHGs.

5 Machine Learning-Based Prediction of Top 10 Polluters

This experiment implements the aforementioned research design. The prediction task is modeled as a binary classification problem in which companies are classified as either a top 10 polluter (1) or a non-top 10 polluter (0) for a given year, branch, and pollution type.

5.1 Data Preparation

The first step in the data preparation process involved selecting PRTR reports and variables essential for the ML model from the PRTR register. The reports were selected based on specific constraints to ensure accurate, up-to-date predictions. Companies were included if they had reported data beyond 2012 and had notifications spanning at least 15 years, including one after 2020. This filtering process resulted in a dataset in which each row represents a company's main annual activity and reported pollution amount.

The dataset was then supplemented with population density data from the German Federal Statistical Office (DStatist) and the official municipality list (June 2024 version). If necessary population numbers were also retrieved from official municipal websites.

Annual branch-specific rankings and top 10 polluter labels were created by comparing reported pollution amounts. Each year, companies ranked from one to ten received a label of one, while others received a label of zero. This process added three columns for GHG emissions: 'AirAmnt' (annual GHG amounts), 'AirRank' (branch-specific rank), and 'AirT10P' (top 10 polluter indicator). Similarly, three columns were added for heavy metal releases into water: 'WaterAmnt', 'WaterRank', and 'WaterT10P'.

5.2 Analysis of Potential Predictor Variables

To improve prediction accuracy, the independent (feature) variables should be strongly correlated with the dependent variables. Using AirAmnt and WaterAmnt as proxies for AirT10P and WaterT10P allowed for more precise correlation analysis. This approach was chosen because numerical variables provide more detailed insights into relationships than binary categories do, particularly given the class imbalance in the dataset.

The following features were analyzed for potential correlations with pollution amounts:

- **Business Sector (NACE Code):** Different sectors have distinct production methods, equipment, and environmental impacts, making this a relevant predictor.
- **Type of Activity (PRTR Key):** Variations in facility activities affect pollution levels, supporting its inclusion as a predictor.
- **Location (Postal Code):** Geographic factors, such as infrastructure, proximity to communities, and environmental regulations, may impact pollution levels.
- **Reporting Year:** Economic conditions and technological progress change annually and may influence pollution amounts.
- **Population Density:** Higher population densities may lead to increased regulatory scrutiny and compliance pressure, which could result in lower pollution levels.
- **Secondary Activities:** A high number of non-main activities with pollution reports indicates a production site with high pollution levels, making this a relevant predictor.
- **Release above limit absolute:** Companies can vary greatly in the amount by which they exceed the regulatory limit. Companies with a large excess amount are more likely to be ranked among the top 10 polluters than companies with a small excess amount.
- **Release above limit relative:** The relative excess amount can also be considered a dependent variable.

The results of ANOVA tests [Co21] indicate that the categorical variables 'Business Sector', 'Type of Activity', and 'Location' significantly influence annual pollution amounts. Pearson correlation tests [Co21] revealed statistically significant but very weak positive correlations for continuous variables (year, population, secondary activities, and release above limit absolute/relative) with the following exceptions. The test between 'Release above limit absolute' and 'AirAmnt' revealed a moderate correlation. Weak positive correlations were obtained for 'Secondary Activities' and the two dependent variables, 'AirAmnt' and 'WaterAmnt'.

5.3 Resampling and Modeling

The analytical dataset contained several columns with missing values, especially for pollution amounts, pollution rank, and top 10 polluter labels. This was because the dataset included reports for all PRTR pollution types, not just the types targeted by the prediction models. Additionally, even within subsets of pollution reports focusing on specific pollution types, years, and industry branches, the data exhibited significant class imbalance. This imbalance occurred because only 10 reports could be labeled '1' to indicate a top polluter, while the rest were labeled '0' to indicate non-top polluters. Consequently, in branches with numerous companies, the proportion of '1' labels was substantially outnumbered by '0' labels.

To ensure a robust foundation for model development, it was necessary to address this imbalance using appropriate data resampling techniques. Therefore, five downsampling steps were applied to the majority class. First, all records from firms that did not submit

any reports after 2017 were removed from the dataset. Second, firms that submitted five or fewer activity reports up to and including 2020 were identified, and all of their corresponding records were excluded from the dataset. Third, eighty percent of reports from 2007 to 2012 containing missing values in any independent variable were randomly selected and removed from the dataset. Fourth, for reports from 2013 to 2018 that had missing values in an independent variable, 40% were randomly selected and removed from the dataset. Fifth, across the entire period from 2007 to 2022, 70% of reports containing missing values in an independent variable were randomly removed from the dataset. In addition to downsampling oversampling was applied using the SMOTE method [Ch02] in order to increase the minority class.

The training and validation of Random Forrest Models for the two classification tasks was performed based on K-fold cross-validation with 5 folds.

5.4 Evaluation – First Experimental Results

The validation results for the two models that predict the binary dependent variables AirT10P and WaterT10P are presented in Table 1. The performance measures reported use common ML indicators: accuracy, precision, recall, F1 score, Cohen’s kappa (CK), and Area Under the Curve (AUC) [RTK24][21].

Variable	Accuracy	Precision	Recall	F1 Score	CK	AUC
AirT10P	0.980	0.981	0.980	0.981	0.829	0.986
WaterT10P	0.978	0.980	0.978	0.978	0.797	0.982

Tab. 1: Performance validation results obtained through k-fold cross-validation

All indicators are close to the maximum possible value of 1.0, except for Cohen’s kappa, which is slightly lower. These performance metrics suggest that the predictions of both models are highly consistent with the true labels. Additionally, the Cohen’s kappa values show that the models’ agreement with the true labels is significantly better than would be expected by chance.

5.5 Model Deployment and Validation based on New Data for the Year 2023

The models were trained using pollution reports from 2007 to 2022. In March 2025, a new version of the PRTR dataset was released that included data for the 2023 reporting year. This update permitted a meaningful out-of-sample evaluation of model performance by comparing the predicted top ten polluters for 2023 with the actual reported outcomes.

To support model deployment, an input dataset was constructed to mirror the structure and features of the training data. Since the models were designed to perform binary classification the input dataset had to include firms with a reasonable likelihood of having reported pollution in 2023. To this end, firms that reported any pollution related to their

primary activity in 2022 were selected. This decision was informed by prior research [TS25], which indicated that firms active in the most recent reporting year are more likely to continue reporting in subsequent years.

The base input dataset was derived from the 2022 subset of the training data and included 5,371 company records. Standard preprocessing steps, were applied to all numerical features of the dataset, consistent with the original training pipeline. Input vectors were constructed using the same set of predictor variables as during training. Static variables, such as 'Business sector', 'Type of activity', and 'Location', retained their 2022 values because they are unlikely to change from year to year. New values were computed for time-sensitive variables, such as 'Secondary Activities', 'Release of GHG into the air above limit abs.', and 'Release of GHG into the air above limit rel.' as the average over the years 2007 to 2022. This strategy was adopted due to the lack of 2023 values and is supported by findings in [TS25], which observed variability in these indicators from year to year. The variable 'Year' was set to '2023' to reflect the prediction target.

The prediction results were compared to the actual top 10 lists of the year 2023. Model performance was evaluated using standard classification metrics: True positives (TP), true negatives (TN), false positives (FP), false negatives (FN), accuracy, and precision. Tables 2 and 3 present the detailed performance indicators for each industrial sector.

Branch	TP	TN	FP	FN	Accuracy	Precision
Waste and wastewater management	7	1998	3	3	0.997	0.7
Chemical industry	7	567	3	3	0.989	0.7
Energy sector	6	226	4	4	0.966	0.6
Intensive livestock farming and aquaculture	0	503	10	0	0.980	0.0
Food industry	0	344	10	0	0.971	0.0
Metal industry	8	985	2	2	0.995	0.8
Mineral processing industry	8	218	2	2	0.982	0.8
Paper and wood industry	1	135	9	6	0.900	0.1
Other industrial sectors	2	285	8	0	0.972	0.2

Tab. 2: Evaluation of predicted Top 10 Polluters for Release of Heavy Metals into the water

Across most sectors, the models achieved high accuracy, often exceeding 95%. However, precision varied significantly between sectors. For instance, the metal and chemical industries yielded high accuracy and precision, indicating robust predictive capability. In contrast, sectors such as intensive livestock farming, the food industry, and other industrial sectors demonstrated low precision. Some cases, such as heavy metal discharge in the food industry, showed a precision of 0%. These results are largely attributable to the class imbalance inherent in the data, where the number of negative cases (non-Top 10 polluters) significantly exceeds the number of positive cases, especially since each branch can contain a maximum of ten Top polluters by definition.

Interestingly, model performance was consistent across pollution types in certain

branches, such as waste and wastewater management and the chemical industry. This suggests similar patterns of pollution behavior and model interpretability. For other branches, substantial variation was observed between the two pollution types.

These outcomes highlight the strengths and limitations of the current model design. While the model is accurate overall, the lower precision in some branches indicates the need for further refinement, potentially through the use of additional sector-specific features, improved class balancing techniques, or ensemble methods.

Branch	TP	TN	FP	FN	Accuracy	Precision
Waste and wastewater management	6	1997	4	4	0.996	0.6
Chemical industry	7	567	3	3	0.989	0.7
Energy sector	8	228	2	2	0.983	0.8
Intensive livestock farming and aquaculture	0	503	10	0	0.980	0.0
Food industry	5	341	5	3	0.977	0.5
Metal industry	9	986	1	1	0.997	0.9
Mineral processing industry	5	215	5	5	0.956	0.5
Paper and wood industry	7	138	3	3	0.960	0.7
Other industrial sectors	3	283	7	2	0.969	0.3

Tab. 3: Evaluation of predicted Top 10 Polluters for Release of GHG into the air

6 Conclusions

This article describes an experiment that uses PRTR data to develop ML models that predict the top 10 polluters based on their releases of GHG into the air and heavy metals into the water. The validation results indicate that both models produce predictions that are consistent with the true labels. The dataset can support the development of additional ML models, including regression models that predict future pollution levels and detect cases warranting site inspections. These developments can advance environmental protection applications. However, it is important to note that ML predictions are probabilistic and depend on high-quality real-world data. Misleading predictions due to data limitations could lead to unjustified accusations, which emphasizes the importance of verifying prediction results, especially when they are shared publicly.

Future work will involve developing prediction models for additional pollution types beyond those addressed in this work. By augmenting the dataset, global rankings across all pollution types can be generated. This expanded dataset may lead to new features, improving ML models or enabling the development of regression models that predict companies' future pollution levels. Furthermore, the analysis will investigate patterns of notifications related to individual companies, business sectors, cities, regions, and federal states. Identifying patterns across different pollution types could reveal interdependencies caused by technical processes, enabling further model refinements.

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